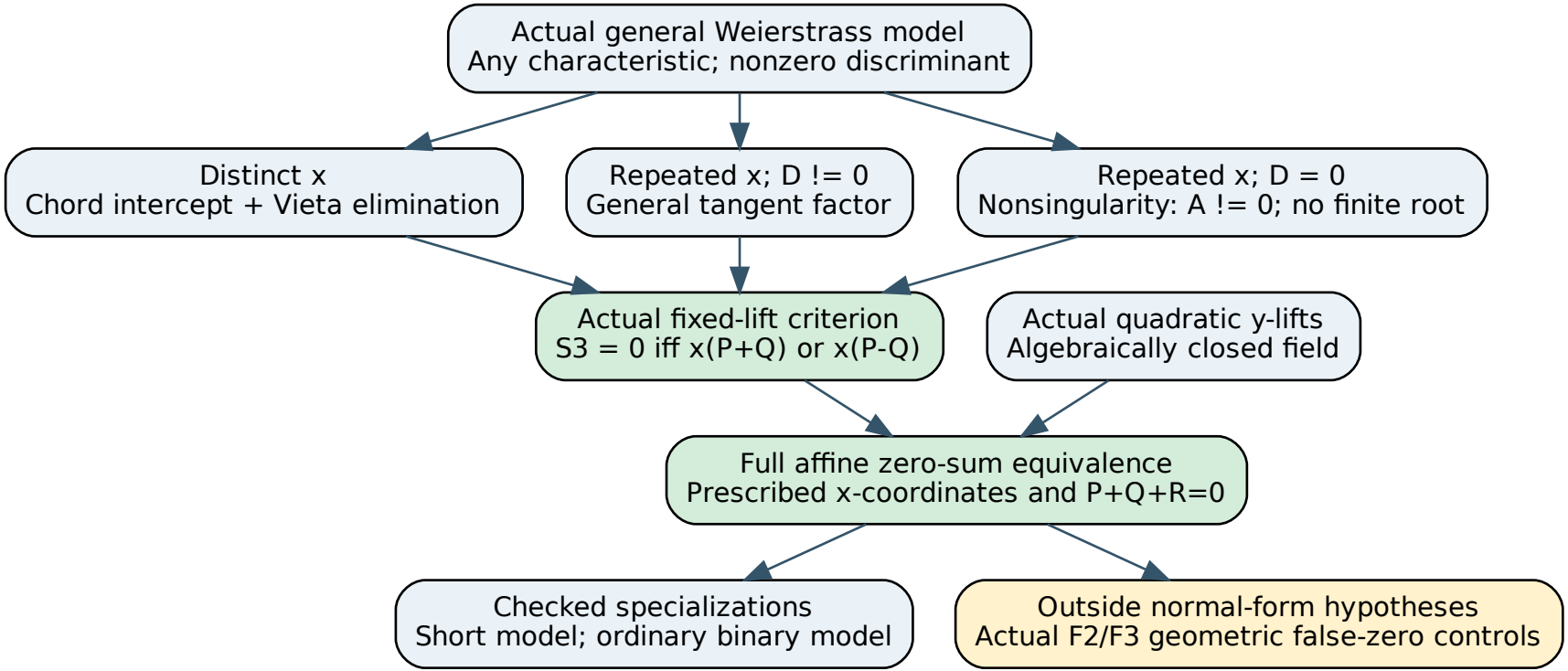
Actual general Weierstrass model
Any characteristic; nonzero discriminant
Distinct x
Chord intercept + Vieta elimination
Repeated x; D != 0
General tangent factor
Repeated x; D = 0
Nonsingularity: A != 0; no finite root
Actual fixed-lift criterion
S3 = 0 iff x(P+Q) or x(P-Q)
Actual quadratic y-lifts
Algebraically closed field
Full affine zero-sum equivalence
Prescribed x-coordinates and P+Q+R=0
Checked specializations
Short model; ordinary binary model
Outside normal-form hypotheses
Actual F2/F3 geometric false-zero controls